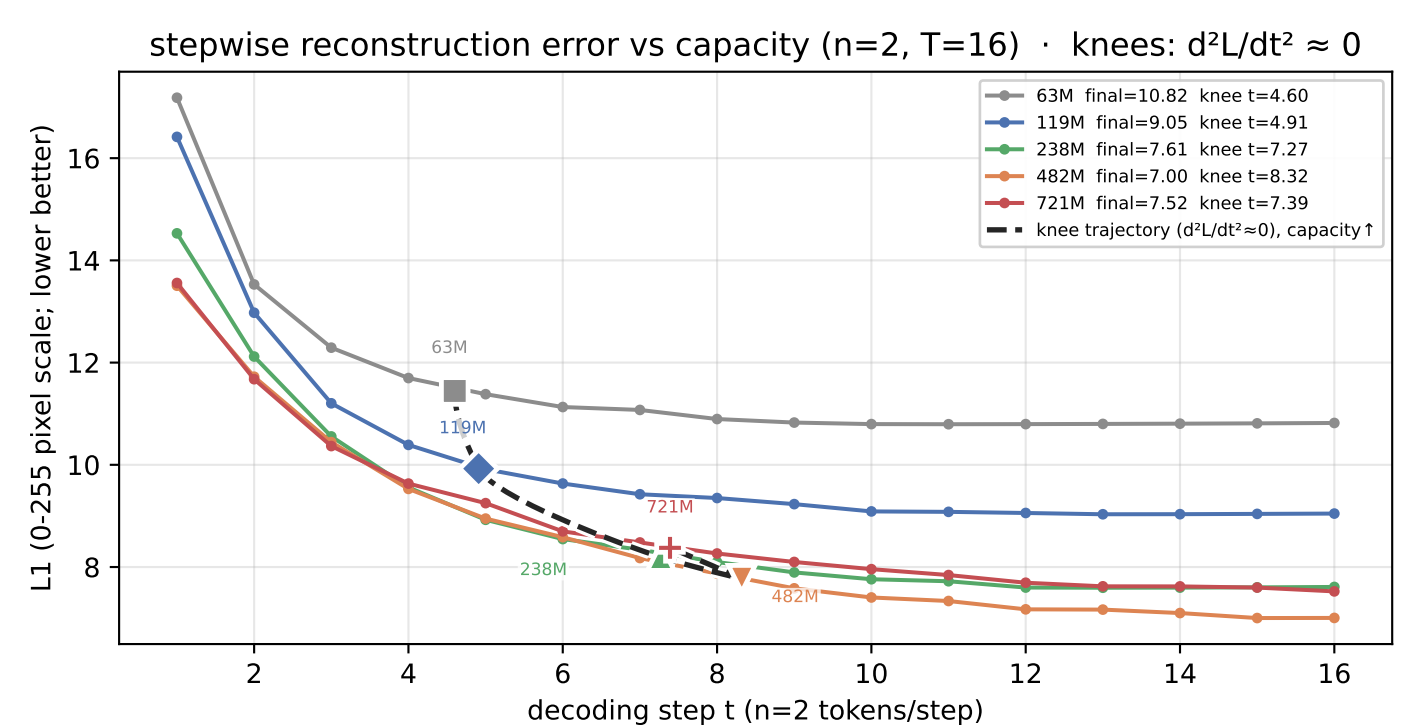

knee = d²L/dt² ≈ 0: first t∈[1.5,15.5] with |d²L/dt²| ≤ 10% of its peak (≈ second derivative decayed to zero).
smooth: Savitzky–Golay (w=5, p=2) → UnivariateSpline (s=0.001·n) → d²L/dt² on a 0.01-step grid; elbow (max|d²L/dt²|) is only a cross-check (t ≈1.5–2.0).
All five curves are monotone convex decays → no strict interior d²L/dt²=0; the 63M baseline has one strict crossing at t ≈13.1 (tail upturn), not marked.